Functional sequences, series and spaces – 2

Series of functions

Homework

13 сентября 2026 года

Theoretical minimum. Weierstrass' test gives uniform and absolute convergence simultaneously and therefore never applies to a conditionally convergent series; there one uses the uniform versions of the Dirichlet and Abel tests, which are Abel summation with the estimates taken uniformly in x. A uniformly convergent series of continuous functions has a continuous sum and can be integrated term by term; to differentiate term by term one needs uniform convergence of the differentiated series, not of the original one. To build a function with prescribed bad points, put a scaled copy of a singularity at each of them: $\sum 2^{-n}\varphi(x - r_n)$ and $\sum\{nx\}/n^2$ are the two standard machines. Finally, a series whose terms have a fixed sign behaves like a numerical series with a parameter, and for it uniform convergence follows from continuity of the sum on a compact set — Dini's theorem in the form for series.

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1. Let (r_n) be an enumeration of the rationals in $[0, 1]$. Prove that $F(x) = \sum_{n\geqslant 1} 2^{-n}|x - r_n|$ is continuous on $[0, 1]$, that F is not differentiable at any rational, and that F is differentiable at almost every point.

2. Prove that $\sum_{n\geqslant 1} \frac{(-1)^n}{n+x}$ converges uniformly on $[0, \infty)$ but not absolutely at any point, and find $\lim_{x\to\infty}$ of the sum together with the first term of its asymptotics.

3. (ISTCiM 2021) Prove that $\sum\limits_{n,m=1}^{N} \frac{(-1)^{n+m}}{n^n m^m (n+m)^2} > 0$ for every positive integer N.

4. Prove that $W(x) = \sum_{n\geqslant 0} a^n \cos(b^n x)$ with $0 < a < 1$, $ab > 1$ is continuous on $\mathbb{R}$ and that the series may not be differentiated term by term; state the condition under which W is nowhere differentiable and say where this is proved (Differentiable functions – 8).

5. Prove that $\zeta(x) = \sum_{n\geqslant 1} n^{-x}$ is infinitely differentiable on $(1, \infty)$, that the differentiated series converges uniformly on $[1 + \delta, \infty)$ but not on $(1, \infty)$, and that $\zeta(x) \sim \frac{1}{x-1}$ as $x \to 1^+$.

6. Find the set of x for which $\sum_{n\geqslant 1} \frac{x^n}{1+x^n}$ converges, prove that the convergence is uniform on compact subsets of $(-1, 1)$, and find the asymptotics of the sum as $x \to 1^-$.

7. Prove Dini's theorem for series: if $u_n \geqslant 0$ are continuous on a compact K and $\sum u_n$ has a continuous sum, then the convergence is uniform. Give a counterexample when the terms are allowed to change sign.

8. Give an example of a series of C^∞ functions which converges uniformly, whose differentiated series converges pointwise everywhere, and for which the sum of the differentiated series is nevertheless not the derivative of the sum.

9. Prove that if $\sum u_n$ converges uniformly on a set E and all u_n are continuous, then it converges uniformly on the closure of E. Deduce that for continuous terms uniform convergence on the rationals of $[0, 1]$ implies uniform convergence on $[0, 1]$.

10. Let u_n be continuous on $[0, 1]$ with $\sum u_n$ converging pointwise to a continuous sum. Prove that for every $\varepsilon > 0$ every subinterval of $[0, 1]$ contains an interval on which the tails are eventually smaller than ε uniformly, and show by an example that the convergence need not be uniform on any subinterval. (Use the Baire category theorem, Continuous functions – 8.)